But The Bot Told Me!

Proposing a Benchmark Concept to Evaluate the Reliability of Large-Language Models as a Tool for Day-to-Day Knowledge Gathering

by

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But The Bot Told Me!

submitted by Anuk Ahangamgoda in partial fulfilment of the requirements of the degree of B.A. Computer Science Honours

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Abstract

While there are several comprehensive benchmarks that evaluate the performance of Large Language Models (LLMs) on specific tasks and domains of knowledge, there remains a lack of a benchmark to measure the reliability of LLMs as a tool for knowledge-gathering across non-professional, everyday domains. This thesis therefore proposes a conceptual benchmark that can evaluate an LLM's ability to answer objective questions reliably within this mundane context. By crafting a novel dataset of prompts across eight distinct knowledge domains and three knowledge classes, the output of an LLM is quantitatively analyzed with respect to key reliability metrics. A secondary dataset, created by converting the prompts into nonsensical counterparts, is used to test if the LLM understands inherently flawed questions. After using a script for automated output generation, an overall reliability score is computed with the key metrics and then given to each tested LLM to create a benchmarking leaderboard. This thesis found that by using the concept benchmark described, the reliability of an LLM in this mentioned context can be successfully evaluated. Further, the research found that variations of a model with fewer parameters or with edge-device optimizations perform less reliably than the original, or larger, models - this was highlighted by the model *gemma3n:e2b* having a reliability score of 57% as an optimized 2-billion parameter model, while *gemma3:4b* had a higher reliability of 61%. The study also showed that the limitations of an LLM can reduce reliability, such as the training date of its internal dataset affecting date-based questions, reducing reliability in domains like *Global news and culture* disproportionately.

Lay Summary

This thesis proposes a method to measure and rank the reliability of answers given by different AI chatbots, and other technology based on Large Language Models (LLMs), when asked questions by non-professionals in day-to-day scenarios. To do so, a list of questions is created, categorized by domains like 'cooking' and types of knowledge like 'myths', and a second list is made using 'nonsensical' versions of questions in the original list. The answers given by a chatbot from the first question list is analyzed according to specific traits, such as whether it contains misinformation. The answers corresponding to the second list are then analyzed to see if the chatbot knows the question itself is nonsensical. The results of the analysis are put through a formula to get an overall score for the chatbot's reliability, and different chatbots can be ranked to provide a benchmark. The thesis found that using the method described, the reliability of a chatbot in this given context can be successfully measured. Further, the study showed that smaller versions of an LLM and versions designed for weaker devices, such as smartphones, are less reliable than the original or bigger, version. The study also found that since LLMs are trained on information up to a particular date, they struggle to reliably answer questions after their training date.

Preface

This thesis was completed by Anuk Ahangamgoda under the supervision of Dr. Adaji Ifeoma in partial fulfillment of the requirements for the Bachelor of Arts with Honors in Computer Science at the University of British Columbia, Okanagan. This includes the research and results analysis conducted, the methodology and pipeline used, the created datasets, and the written work.

Generative artificial intelligence was used solely as the subject of the research conducted, and not for the drafting, or preparation, of the thesis.

This work has not been previously published.

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Chapter 1

Introduction

1.1 Background

A Large Language Model (LLM) is a form of computational model that is able to execute Natural Language Processing (NLP) tasks, which are tasks that involve processing a language organically, through the use of a neural network. LLMs can be implemented as generative artificial intelligence (AI, or GenAI) models [BS23], capable of generating text that can often mimic human language patterns effectively [KM21] and thereby making this technology a popular base for chatbot services such as OpenAI's ChatGPT. These LLM-based services can be used to execute NLP tasks through platforms that are easy to use and accessible [MH25], allowing non-professional users to leverage this technology for personal-use purposes such as creative text generation or simply asking questions to get answers.

Generally, LLMs are evaluated on its *perplexity* - a measure of the model's ability to predict a certain dataset. However, in order to evaluate the performance of LLMs in NLP tasks rather than only dataset prediction, a plethora of standardized benchmarks have been developed, many of which can be found on websites such as *llm-stats.com*. To provide a deeper understanding of an LLM's capabilities, these benchmarks often constrain themselves to specific tasks, such as the ability to write software code [JYW+24] or answer high-level specialist questions [PGHL25]. Further, most benchmarks also constrain themselves to specific fields, or groups of fields, of study, such as biology [TGZ+24] or creative writing [KM21]. The results of benchmarks are valuable to LLM developers and users alike since they highlight the strengths and weaknesses of the model and allow a leader-board to be formed, making it easier to draw comparisons between models.

For tasks that require a degree of reliability when generating text, such as when summarizing a passage or obtaining information accurately, having a benchmark that can evaluate an LLM's reliability is critical. Benchmarks for these types of NLP tasks act as a significant aid when selecting a model since LLMs are prone to a phenomenon known as 'hallucinations', where LLM-generated texts contain instances of fabricated data, flawed logic and

reasoning, or biases [PDY+25] that can degrade the model's reliability. For example, a summary that contains additional fabricated information cannot be used to represent the original text, thus a benchmark for summary reliability can be used to find which models perform better than others.

1.2 Motivation

There exists a significant number of benchmarks for LLMs, with *llm-stats.com* presenting over 480 of them. Despite this wide variety, there still remain certain tasks and fields of study that do not have associated benchmarks, one of which being the evaluation of an LLM's ability to produce reliable output for knowledge-seeking queries in mundane domains of knowledge. An example of such a query could be "give me a recipe for sourdough bread" - here, the LLM is given a knowledge-seeking query that does not require a specialist to understand (seeking a sourdough bread recipe) in a mundane domain of knowledge (cooking in a home environment). To elaborate, the act of knowledge-seeking, or knowledge-gathering, is where an individual attempts to obtain data or information. Mundane domains of knowledge are topics that require non-specialist understanding, such as non-professional work or hobbies, and can exist in a day-to-day environment. Reliable output, in this context, is output that allows for successful knowledge seeking; output is reliable when it is objective, factual, and does not lack key data.

Understanding an LLMs ability to provide reliable output in mundane domains of knowledge is critical when exploring the use of LLM-based chatbot services as a tool for gaining knowledge. With the use of LLM-based chatbots rising [BYE+24], their use as a knowledge-seeking tool, instead of similar tools such as search engines or encyclopedias, may rise as well. Unlike alternate tools, these services are prone to unique drawbacks based on their underlying technology, one notable example being LLM hallucinations - since a hallucinated output cannot be reliable knowledge, it is detrimental to this use-case. Further, the increasing usage of LLM-based chatbot services outside of work-related contexts [CCD+25] indicate that a notable portion of queries would be for non-professional and day-to-day use. This upward trend in the mundane and non-professional use of LLM-based chatbots therefore emphasizes the need to explore their reliability as a tool for knowledge-seeking in mundane contexts.

To provide a solution for measuring the performance of LLMs within this specific context, this thesis proposes a concept for an evaluative benchmark.

The conceptual benchmark uses a novel dataset consisting of prompts categorized into eight mundane domains to be fed into the model being tested: *cooking, personal finance, general knowledge, career advice, academia, computer programming, personal health and fitness,* and *global news*. Additionally, this corpus of prompts is classified into three classes per domain: *general*, *myth*, and *niche*. The model's output for each prompt is then analyzed according to key metrics: whether it *contains misinformation*, whether it *provides sources*, and whether it *lacks key information*. A secondary dataset is created through converting the *general* prompts into nonsensical forms to test if the model understands that the input itself is flawed. These four metrics each have a determined value based on the trifecta *TRUE, FALSE, PARTIAL*. This data can then be used to quantitatively analyse the model and provide an overall 'reliability score' using a weighted formula, as well as provide scores per knowledge-domain and knowledge-class. A quantitative result allows different models to be directly compared with each other, thereby providing an effective means of benchmarking. For this thesis, three LLMs from the *Gemma 3* family of models (*gemma3:4b, gemma3n:e4b, gemma3n:e2b*) were used to verify the methodology and gather results.

1.3 Research Questions

This thesis is therefore guided by the following research questions:

RQ I. *How can the reliability of a large-language model be measured when used in mundane non-professional contexts as a means of answering objective questions?*

RQ II. *What are the limitations of a large-language model that can affect its reliability within this given context?*

RQ III. *How will variations of the same model change its reliability within this given context?*

1.4 Contributions

This paper contributes the following:

- A novel benchmark concept and dataset pair to evaluate LLM reliability as a tool for non-professional, mundane knowledge-gathering.
- A pipeline to automate the benchmarking process.
- Benchmarking results and findings from the tested LLMs.

Chapter 2

Related Works

A significant amount of vital research on the evaluation of LLMs through datasets, frameworks and benchmarks exist, several of which are similar to the research discussed in this paper.

BoolQ, introduced by Clark et al. [CLC+19], is a benchmark designed to test an LLM's ability to respond specifically to yes/no questions that occur naturally. The study devised a novel dataset that contained questions that could only be answered with a 'yes' or 'no'. Similar to this thesis, the questions in BoolQ required a ground truth answer and did not require specialist knowledge to parse, categorized into topics such as *entertainment media* or seeking general world knowledge. The dataset, however, did not explicitly seek to contain only questions across mundane topics, and were limited to yes/no questions as was the goal of the study. The benchmark is thus not an appropriate measure of knowledge-gathering, since knowledge gathering often requires output beyond 'yes' or 'no'.

SimpleQA, a similar benchmark, was developed by Wei et al. [WKC+24] to evaluate if LLMs can generate answers that are factually correct. The benchmark used a novel dataset composed of short prompts that aimed to gather facts. The prompts were constrained by vital criteria, such as requiring answers that do not change over time, and spread across topics such as *politics* and *art*, albeit unevenly distributed. Notably, one of the prompt constraints required the question to be challenging, thereby distinguishing the questions from the trivial knowledge-seeking seen in more mundane topics and may require specialist knowledge to answer. The benchmark is therefore unrepresentative of non-professional, mundane knowledge-gathering.

Natural Questions, a comprehensive dataset designed by Kwiatkowski et al. [KPR+19], consists of knowledge-seeking question quadruplets (*question, WikiPedia page, long answer, short answer*) for benchmarking open domain question-answer (QA) systems. The questions within the corpus are entirely human created, since they are taken from Google search queries, giving the question set stronger validity as a means of mimicking questions from real users of a system. Due to the nature of open domain systems, however, the dataset does not constrain questions to any specific set of topics or diffi-

culty, resulting in questions that could be from professional contexts, unlike what this thesis proposes.

MMLU (Massive Multitask Language Understanding), by Hendrycks et al. [HBB+21], proposed a benchmark to evaluate Transformer models on their ability to acquire knowledge during the pretraining process on subjects that humans can learn, giving the benchmark some similarity with methods that evaluate humans. The question set, while containing simpler elementary-level questions, also includes complex questions that require a professional degree of understanding, making it ineffective for the purpose of evaluating knowledge-gathering where a specialist is not needed.

Other similar benchmarks include **TriviaQA**, introduced by Joshi et al. [JCWZ17], which presents a large-scale benchmarking dataset of over 650,000 (often challenging) trivia-style questions that aim to test knowledge rather than seek knowledge; **PIQA**, introduced by Bisk et al. [BZB+19], which brings forth a dataset specifically for benchmarking a model's commonsense reasoning in physical spaces; and **MMLU-Pro**, a dataset by Wang et al. [WMZ+24] which expands on the **MMLU** benchmark with challenging questions that require reasoning, whilst culling existing trivial questions.

Benchmarks such as *BoolQ, MMLU, TriviaQA* and others therefore do not truly fill the gap towards gaining insight on how reliable an LLM can be when used as a tool for mundane, non-professional knowledge gathering due to their underlying methodologies, such as *BoolQ*'s yes/no question dataset being inappropriate for knowledge gathering or *TriviaQA*'s dataset being limited to trivia-style quiz questions rather than authentic knowledge-gathering questions. The benchmark concept proposed in this thesis can thus ensure this gap is filled, since it will directly evaluate an LLM on a dataset designed to be representative of non-professional, mundane questions that have the aim of knowledge-gathering and not knowledge-testing, as well as being balanced across eight specific topics to ensure a meaningful spread of information.

Chapter 3

Methodology

The proposed benchmark consists of three distinct sections: the dataset of input prompts, the software pipeline used to collect the output of the tested LLMs, and the quantitative analysis of the output for each tested LLM. In the first section, a novel dataset is created, composed of a variety of categorized prompts. Each prompt has accompanying data, being the 'ground truth' statement, 'key facts' statement, and 'search-engine query equivalent' statement, as well as having a classification for the prompt's domain and knowledge type. Then, a secondary dataset is created by converting specific prompts from the primary dataset into 'nonsensical' forms. In the second section, the selected LLM is fed both datasets using a pipeline built with Python and Ollama. The pipeline outputs two files, one for each dataset, and each containing the LLM's output per prompt. For the third and final section, the output files are manually analyzed with respect to each prompt's accompanying data, allowing the LLM to be given a reliability score computed using the analysis results.

This process can then be repeated across different models, allowing a comparative analysis of each model's reliability based on their reliability scores.

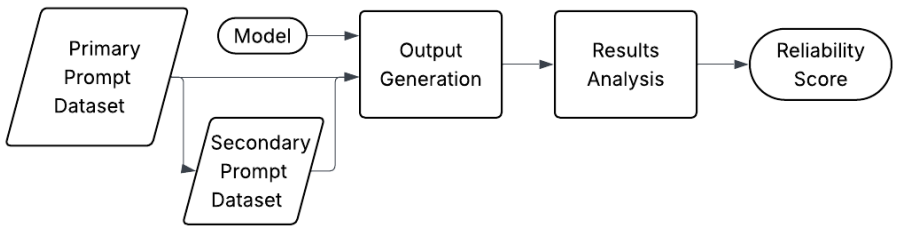


Figure 3.1: Overall evaluation pipeline.

Table 3.1: Example prompts for the domain *cooking*.

Domain	Prompt Type	LLM Prompt
Cooking	Day-to-day	Could you give me a recipe for sourdough bread?
	Niche	Could you give me a recipe for 'wattakka kalu pol'?
	Myth	Why is MSG harmful?

3.1 Dataset Creation

Datasets allow a model's performance to be bench-marked through comparing the model's output with an expected output, based on particular metrics. Generally, the dataset would contain the data to be input into the model, as well as its corresponding 'ground truth' and any other necessary information for evaluation. In this concept benchmark, the input data are questions that aim to gather knowledge by non-professionals in mundane environments, thus allowing the model's performance in this particular task to be evaluated. All data in the datasets used were organically created, rather than synthesized, to strengthen their validity as genuine, real-world questions.

3.1.1 Primary Prompt Dataset

The core dataset consists of a collection of unique prompts, each designed for the purpose of this benchmark. The prompts are engineered according to the following constraints:

- **Knowledge-Gathering**: the prompt must be a task with the goal of receiving 'knowledge' from the LLM.
- **Objectionability**: the prompt must request an output that can be objectively verified.
- **Future Data**: the prompt must not request an output that cannot be objectively answered by March 2026.
- **Non-Specialist**: the prompt must not require specialist knowledge to be understood.

– **Search-Engine Results**: the prompt must request knowledge that could otherwise be obtained by websites listed within the 'first page' of the Google search engine.

The constraints have been selected in order to constrain the benchmark to its intended evaluation case (knowledge-seeking in mundane domains) and allow the LLM to respond objectively with existing data at the time of evaluation.

Each prompt must also be assigned one knowledge domain and one knowledge class, as well as a brief justification for the assignment:

– **Knowledge Domains**: *cooking, personal finance, general knowledge, career advice, academia, computer programming, personal health and fitness, and global news*

– **Knowledge Classes**: *general, niche, myth*

The domains chosen encapsulate the various fields that an individual may seek knowledge for in non-professional or hobby settings. The classes chosen provide a more comprehensive evaluation, since they logically separate contexts irrespective of mundanity (whether the question is only asked within a certain culture, thereby making it niche; whether the question has commonly accepted incorrect answers, thereby making it a myth). By spreading prompts across these domains and classes, the dataset can be a balanced representation of the overarching task.

To accommodate the knowledge classes, each class carries additional prompt engineering constraints:

– **General**: The prompt must reasonably mimic a question that could be answered without additional cultural or specialist contexts.

– **Niche**: The prompt must reasonably mimic a question that is within a niche field of knowledge, without being technical.

– **Myth**: The prompt must reasonably mimic a question where the objectionably true answer is superseded by a myth.

The prompt is then reviewed by a domain expert to verify that it meets the constraint criteria and confirm that the prompt mimics an attempt at mundane knowledge-seeking. This reinforces the validity of the prompts. Once reviewed, the prompt is given additional accompanying data to be used during the final output analysis section:

Table 3.2: Example of *bad prompt* conversion.

Original Prompt	What exercises are good for back pain?
Converted Prompt	What exercises can treat backpain according to Faust's Medical Journal?
Conversion Technique	Plausible Non-Existent Framework (Faust's Medical Journal does not exist)

- **Ground Truth**: the objectionably true answer for the prompt.
- **Key Facts**: the data points that must be present in the output in order for it to be considered a complete response.
- **Search Engine Equivalent**: the key terms in the prompt that can be queried by a search engine for an equivalent attempt at knowledge-gathering.

This accompanying data will allow a model's output to be analyzed with respect to certain metrics for a deeper, more nuanced analysis.

For this thesis, the primary dataset contains five prompts per domain, with three prompts being *general* and one additional prompt for *niche* and *myth* each, for a total of 40 prompts across all domains. This allows for an equally distributed representation of all eight domains, and an intuitive spread of classes (there will be more *general* prompts than *niche* and *myth* prompts, since the latter two knowledge classes occur less frequently in knowledge-gathering).

3.1.2 Secondary Prompt Dataset

The secondary dataset consists of 'nonsensical' forms of the *general* prompts in the primary dataset. The 'nonsensical' prompts are labeled as *bad prompt* equivalents, and are used to test if the model is able to understand that the input itself is flawed and still provide a reliable output. This helps account for cases in knowledge-gathering where the question being asked is inherently wrong, and therefore cannot be answered. Since this involves changing the prompt itself rather than observe a model's output, a separate dataset was implemented to allow for a logical separation.

The techniques used to convert prompts into *bad prompts* were adapted

from the guide used in the benchmark *Bullshit Bench* [1], since it provides a comprehensive framework for drafting nonsensical prompts and is backed by many contributors.

For this thesis, the secondary dataset contains three prompts per domain, for a total of 24 prompts, and a grand total of 64 prompts across both datasets.

3.2 Output Generation

A pipeline was developed using Python and Ollama, and integrated via the Ollama Python library, to automate the process of feeding prompts to a model and collecting the output.

Docker is used to host an instance of Ollama, allowing for better reproducibility and consistency in environments. Python was chosen due to its versatility and compatibility with handling deep-learning and big-data systems. Ollama was selected due to its ease of use for hosting LLMs in local development environments, and its integration with Python through the dedicated Ollama Python library. CSV files were used to feed prompts to the Python script, as well as save model outputs, due to their compatibility with Python via the CSV library and their human readability, which is useful for the manual verification of data.

The script begins with specifying a list of models to be tested. For this study, the models tested were *gemma3:4b, gemma3n:e2b* and *gemma3n:e4b* since these models are open-source and were able to run locally on the test system at no additional costs. Further, these models are variations of the *Gemma 3* family of LLMs, where *gemma3:4b* indicates a 4-billion parameter model, *gemma3n:e4b* indicates a 4-billion parameter model for edge-devices, and *gemma3n:e2b* indicates a 2-billion parameter model for edge-devices. Using these variants of the same model family (*Gemma 3*) allows the study to investigate how parameter count and edge-device optimizations can affect model reliability in the context of this thesis.

Next, the script initializes the Ollama API via the Ollama Python library, and iterates through each model in the list. For each iteration, the model is pulled into the test machine to be run locally and then iterates through the list of prompts from a particular dataset without any accompanying information or additional prompts. The API allows for conversation history to be disabled, preventing previous prompts and outputs from

[1] petergpt/bullshit-benchmark: BullshitBench measures whether AI models challenge nonsensical prompts instead of confidently answering them, created by Peter Gostev.

breaching onto the context of the next. A CSV file is created under the name of the model and dataset, appending each output as a new field. After the prompt list from the primary dataset is exhausted, the secondary dataset is then utilized the same way, after which the model is removed from memory to free system resources for the next model to be loaded. After all the models have been run through the script, a collection of CSV files are ready for output analysis, with each model having two accompanying output files corresponding to the primary and secondary datasets. This process thus allows for the automation of output generation for the specified models, reducing manual labour significantly without affecting the validity of the methodology.

3.3 Output Analysis

3.3.1 Primary Dataset Analysis

The analysis process starts with taking each prompt's corresponding output from a particular CSV file and dividing it into individual given statements. If any statement contains information that can be proven false, the output is flagged *TRUE* for 'Contains Misinformation'. Similarly, if at least one piece of key information noted alongside the prompt is missing it would be flagged *TRUE* for 'Misses Key Information'. If a source is provided, it would be flagged *FALSE* for 'Does Not Contain Sources', but could also be flagged *PARTIAL* if sources are provided that do not directly verify factoids. The output is also compared with the ground truth and assigned a *TRUE*, *FALSE* or *PARTIAL* flag to indicate if the output is correct, incorrect, or partially correct overall. These four key metrics allow for each output to be analyzed with more nuance than simply 'correct or incorrect', since it provides a deeper perspective on how well the LLM failed or succeeded at each prompt.

3.3.2 Secondary Dataset Analysis

Unlike the primary dataset, here each prompt's corresponding output is taken and analyzed without statement division. Since this dataset only contains *bad prompts*, the output is analyzed as a whole instead to determine if the model understood the nonsensical prompt and provided a reliable (*TRUE*) answer, did not understand (*FALSE*), or indicated that the question could be flawed but attempted to answer regardless (*PARTIAL*). This provides representation for edge-cases in knowledge-gathering where

the question is inherently flawed, and therefore will never have a reliable answer without the questions having to change, presenting a fifth key metric.

3.3.3 Reliability Scores

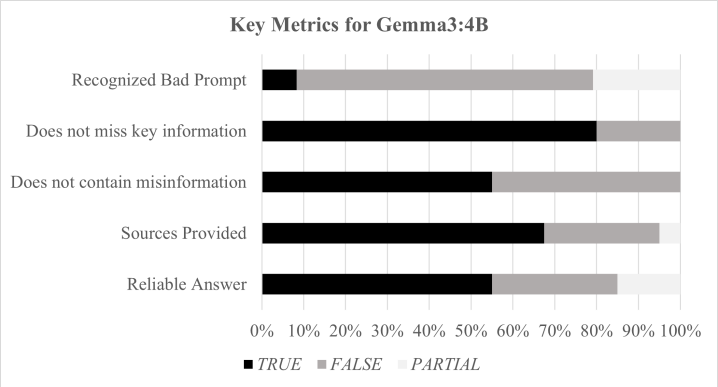


Figure 3.2: Metrics for the model *gemma3:4b*

The overall reliability score is calculated using the weighted average formula:

$$\frac{(\lambda_1 X + \lambda_2 Y + \lambda_3 Z + \lambda_4 P + \lambda_5 Q)}{sum_{i=1}^{5}(\lambda_i + \lambda_{i-1})}$$

where *X, Y, Z, P, Q* represent the five metrics *Reliable answer, Sources provided, Does not contain misinformation, Does not miss key information* and *Understood nonsensical prompt* respectively, and $\sum_{i=1}^{5}(\lambda_i + \lambda_{i-1})$ represents the sum of all weights. For this study, the weights used were $\{4, 2, 4, 2, 1\}$, where *Reliable answer* was given the most weight. A weighted average allows metrics to be given importance, where more important metrics have a higher weight (such as $\lambda_1 = 4$ for X, since *Reliable answer* is the most significant metric when evaluating reliability). The final reliability score will therefore be more accurate than if an unweighted average was used.

To calculate the value of the metrics themselves, the sum was tallied for each prompt and each metric, with *true*=1, *false*=0 and *partial*=0.5, and divided by the total count of prompts, allowing each knowledge-domain and knowledge-class to be assigned scores. This allows each model's reliability to be interpreted with detail beyond a single reliability score.

The final output analysis was done manually to prevent hallucinations in neural-network based analysis methods from degrading validity.

Chapter 4

Results

The models *gemma3:4b, gemma3n:e2b* and *gemma3n:e4b* were benchmarked using the described methodology.

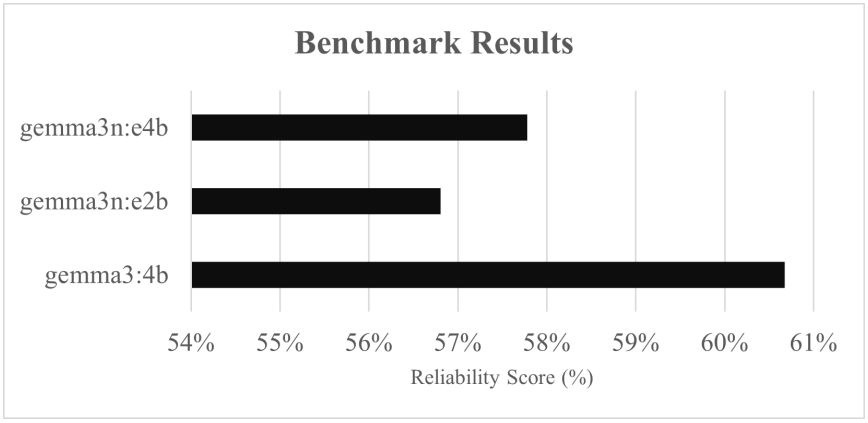


Figure 4.1: Benchmark results for tested *Gemma 3* family of models.

Figure 4.1 and table 4.1 show the benchmark leaderboard, with *gemma3:4b* achieving the highest reliability score of the three, at over 60%, and *gemma3n:e2b* having the lowest score at 57%. The edge-device optimized models *gemma3n:e2b* and *e4b* performed worse than the non-optimized *gemma3:4b* model; the model with the lower parameter count (*e2b* at 2-billion parameters) performed worse against the other two models' 4-billion parameter count.

Figure 4.2 shows the same benchmark results by each of the five key

Table 4.1: Benchmark leaderboard for tested models.

Model	Reliability Score
gemma3:4b	61%
gemma3n:e2b	57%
gemma3n:e4b	58%

metrics, rather than the overall reliability score. All three models perform the worst when understanding if a question is nonsensical, and perform the best when not missing key information in their output. The two edge-device optimized models performed significantly worse when providing sources, at less than 30% for both, as compared to the non-optimized model at 70%. All three models performed about the same at providing a reliable answer, with the edge-device models performing slightly better at near 70% compared to the non-optimized model at over 60%, and in avoiding misinformation, with all three at about 55%.

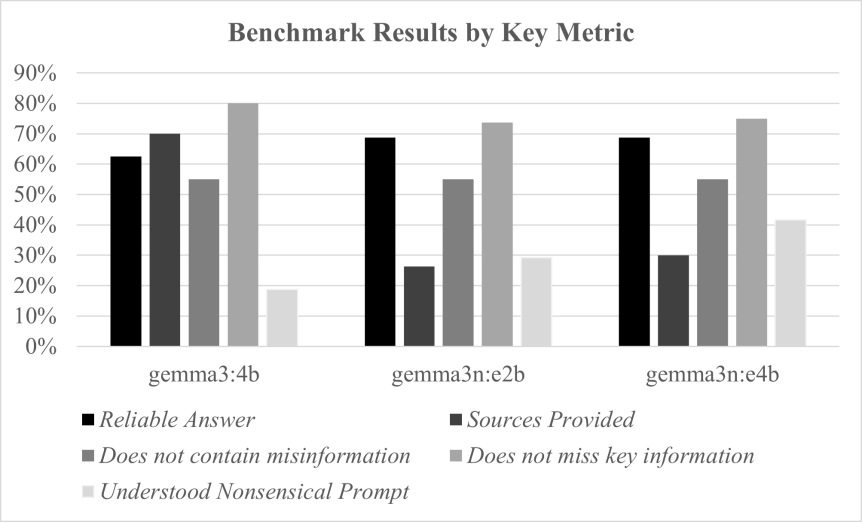


Figure 4.2: Benchmark results by key metric.

Since the methodology allows for deeper understanding of a model, we can investigate results further by taking *gemma3:4b* as an example.

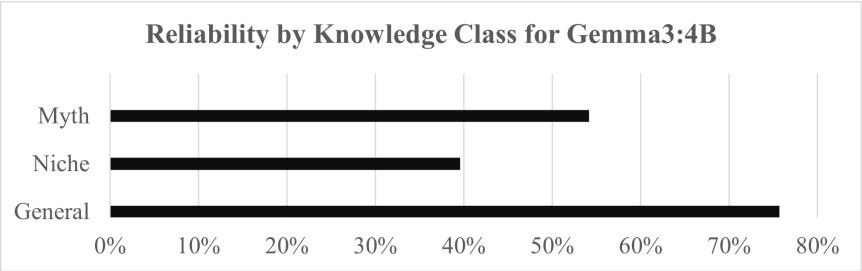


Figure 4.3: *gemma3:4b* reliabilities by knowledge-class.

Figure 4.3 shows that the model *gemma3:4b* is least reliable at *niche* questions, at less than 40% reliability, compared to *general* questions at over 75% reliability. *Myth* questions lay in the middle, at about 55% reliability. Likewise, the model's most reliable knowledge-domain was *programming*, at 90%, while *Global News and Culture* proved the least reliable, at just over 30% reliability.

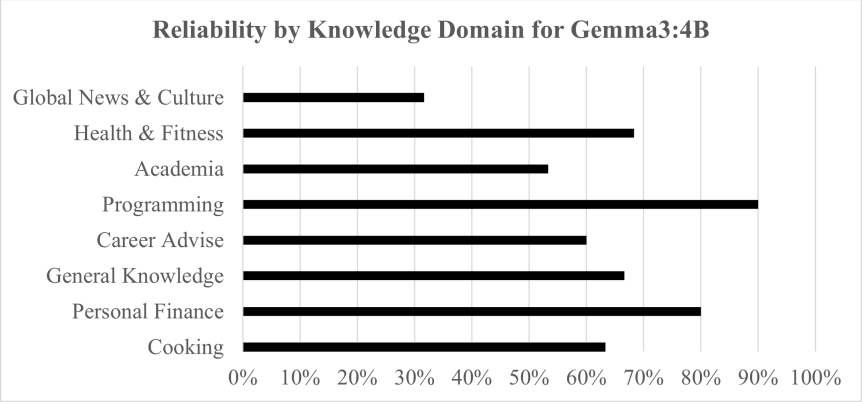


Figure 4.4: *gemma3:4b* reliabilities by knowledge-domain.

Appendix figures A.1 and A.2 breakdown *gemma3:4b*'s output further into each key metric for knowledge-domains and knowledge-classes, allowing strengths and weaknesses in reliability to be identified at more specificity. The figures indicate that, for example, all three knowledge classes performed equally in not missing key information, but *niche* prompts performed extremely poorly in the metrics *reliable answer* and *does not contain misinformation*. The model almost always provided a source when in the domain of *personal finance*, but provided sources less frequently in other domains, such as *cooking*. These specific findings were reflected more broadly in the previous figures 4.4 and 4.3, but analyzing each key metric aids greatly in locating how exactly the model failed or succeeded in its reliability.

Chapter 5

Discussion

5.1 Summary of Key Findings

This thesis introduces a novel dataset, secondary dataset, and methodology to be used for benchmarking the performance of large-language models, specifically for a benchmark concept evaluating their reliability as a tool for mundane, non-professional knowledge-gathering. By benchmarking three versions of the model *Gemma 3*, and guided by the research questions listed in section 1.3, the findings are as follows:

5.1.1 Research Question I

RQ I asks how LLM reliability can be measured in the mentioned context. This can be done through the use of the concept benchmark described in this thesis. As shown in the results, an LLM can be successfully assigned an overall reliability score, expressed as a percentage. The benchmark also allows for more comprehensive evaluation through the use of the five key metrics, knowledge-domains, and knowledge classes, thereby forming a thorough tool for reliability measuring. Further, higher parameter counts and larger models generally result in better performance in benchmarks, as shown in Wang et al.'s 2025 study [WLL+25]. Table 4.1 shows that this pattern stays true in this benchmark concept as well, reinforcing its validity as a successful measure of LLM reliability. **RQ I** can thus be successfully answered through the use of the described concept benchmark.

5.1.2 Research Question II

RQ II explores what limitations of LLMs can affect their reliability in the mentioned context. Using the analysis of *gemma3:4b*'s results, the difference in reliability between the three knowledge-classes is highlighted. The low reliability in *niche* prompts is likely due to the comparative lack of model training data in niche topics, particularly in the English language, as well as myths being conflated with truths in training data when it comes to *myth*

prompts. The model also shows reliability differences between knowledge-domains. A possible reason for the gap in reliability between *Global News and Culture* and the other domains could be the training date of the model - whilst the prompts followed a cutoff date of the March of 2026 as a core constraint, the model's training date was the November of 2023, resulting in a disparity of over two years. This resulted in the model unable to answer questions about events that occurred after its training date, thereby giving a disproportionate amount of unreliable answers as compared to other domains which rely less on date-based queries. These limitations in training data carry over to LLMs trained with it, ultimately causing an LLM's reliability to be affected.

5.1.3 Research Question III

RQ III aims to find out if variations of the same model can affect its reliability in the mentioned context. The final results indicate that the edge-device optimized models were less reliable than the original model, and that the model with the lowest parameter count performed the worst. When investigated deeper, a notable cause of this difference for edge-device optimized models could be that they performed extremely poorly when it came to providing sources, while performing equally in most other metrics. This may hint that the optimizations made result in them being less likely to provide sources. Interestingly, both edge-device optimized models outperformed the original model when understanding if a prompt is nonsensical - this could be investigated further to identify if sycophancy or blind agreement is reduced during edge-device optimizations, thereby being a possible cause for this improvement. These findings therefore answer **RQ III** by illustrating that reliability in answers can be negatively affected by LLM variations, such as lower parameter counts or edge-device optimizations.

5.2 Limitations

- Due to hardware limitations and minimizing financial expenditure, the models that could be bench-marked were restricted to models that could run locally on the development machine at no additional costs.

- Only three models were run through the output analysis stage, since manual analysis was time-consuming. Further, analysis was done by one person, thus human error may be present in the results. For the

same reasons, models were not run multiple times to find an average of results, thus outlying data may not have been identified.

- The size of the datasets used for benchmarking are extremely small as a result of creating organic data instead of synthesizing data, thus it may not be adequately comprehensive for meaningful benchmark results.
- Knowledge domains and classes are limited to those selected and may not be a proper representation of real-world usage.
- The techniques used to form *bad prompts* were adapted from a benchmark that is not formally reviewed and published, affecting its validity and thus the validity of the secondary dataset.

5.3 Future Work

- The strongest improvement that could be made is implementing automation for the output analysis stage. For example, using fine-tuned models in a judge panel voting system in place of manual analysis could reduce human error and save time and labour in a way that retains validity.
- Using a much larger dataset and more knowledge-domains and knowledge-classes can help make the benchmark much more representative of real-world usage and generalizable.
- When comparing variations of a model to investigate how that can affect reliability, it may be insightful to compare between generations of a model (such as between *Gemma 2* and *Gemma 3*) as well.
- Forming *bad prompts* using a technique that has stronger validity may help in creating a stronger secondary dataset.
- The models tested do not possess 'web search' functionality, and were all tested at the default model temperature. Future work could include these factors to explore how they may affect model reliability.

Chapter 6

Conclusion

This thesis aims to address how the reliability of large-language models in mundane, non-professional knowledge-gathering tasks can be evaluated, since there is a gap in the literature for a method of evaluation in this context. Through the use of a novel dataset pair, software pipeline, and methodology, a concept benchmark is designed for the purpose of quantitatively measuring reliability in the mentioned context, allowing for LLMs to be evaluated and compared. This concept benchmark also allowed for an exploration into factors that could affect reliability, such as model variations and model limitations, as well as present the benchmarking results for three models in the *Gemma 3* family. The results find that the concept benchmark can successfully serve as a tool for evaluation, that limitations in training data, such as training date, in LLMs can affect their reliability, and that model variations with lower parameter counts or edge-device optimizations perform worse in terms of reliability.

Bibliography

[BS23] Leonardo Banh and Gero Strobel. Generative artificial intelligence. *Electronic Markets*, 33(1):63, December 2023. → pages 1

[BYE+24] Md. Asaduzzaman Babu, Kazi Md. Yusuf, Lima Nasrin Eni, Shekh Md. Sahiduj Jaman, and Mst. Rasna Sharmin. ChatGPT and generation 'Z': A study on the usage rates of ChatGPT. *Social Sciences & Humanities Open*, 10:101163, January 2024. → pages 2

[BZB+19] Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. PIQA: Reasoning about Physical Commonsense in Natural Language, November 2019. → pages 5

[CCD+25] Aaron Chatterji, Thomas Cunningham, David J. Deming, Zoe Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman. How People Use ChatGPT, September 2025. → pages 2

[CLC+19] Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. BoolQ: Exploring the Surprising Difficulty of Natural Yes/No Questions, May 2019. → pages 4

[HBB+21] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring Massive Multitask Language Understanding, January 2021. arXiv:2009.03300 [cs]. → pages 5

[JCWZ17] Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension, May 2017. arXiv:1705.03551 [cs]. → pages 5

[JYW+24] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench:

Can Language Models Resolve Real-World GitHub Issues?, November 2024. arXiv:2310.06770 [cs]. → pages 1

[KM21] Nils Köbis and Luca D. Mossink. Artificial intelligence versus Maya Angelou: Experimental evidence that people cannot differentiate AI-generated from human-written poetry. *Computers in Human Behavior*, 114:106553, January 2021. → pages 1

[KPR+19] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. Natural Questions: A Benchmark for Question Answering Research. *Transactions of the Association for Computational Linguistics*, 7:452–466, 2019. → pages 4

[MH25] Il Bong Mun and Kyung-Ho Hwang. Exploring the Influence of Prompt Self-Efficacy: Accurate and Customized Information, Perceived Ease of Use, Satisfaction, and Continuance Intention to Use ChatGPT. *International Journal of Human–Computer Interaction*, 41(22):13952–13963, November 2025. _eprint: https://doi.org/10.1080/10447318.2025.2480823. → pages 1

[PDY+25] Parth Pulkundwar, Vivek Dhanawade, Rohit Yadav, Minal Sonkar, Medha Asurlekar, and Sarita Rathod. A Concise Review of Hallucinations in LLMs and their Mitigation, 2025. Version Number: 1. → pages 2

[PGHL25] Long Phan, Alice Gatti, Ziwen Han, and Nathaniel Li. Humanity's Last Exam. *SuperIntelligence - Robotics - Safety & Alignment*, 2(1), March 2025. → pages 1

[TGZ+24] Minyang Tian, Luyu Gao, Shizhuo Dylan Zhang, Xinan Chen, Cunwei Fan, Xuefei Guo, Roland Haas, Pan Ji, Kittithat Krongchon, Yao Li, Shengyan Liu, Di Luo, Yutao Ma, Hao Tong, Kha Trinh, Chenyu Tian, Zihan Wang, Bohao Wu, Yanyu Xiong, Shengzhu Yin, Minhui Zhu, Kilian Lieret, Yanxin Lu, Genglin Liu, Yufeng Du, Tianhua Tao, Ofir Press, Jamie Callan, Eliu Huerta, and Hao Peng. SciCode: A Research Coding Bench-

mark Curated by Scientists, July 2024. arXiv:2407.13168 [cs]. → pages 1

[WKC+24] Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese, John Schulman, and William Fedus. Measuring short-form factuality in large language models, November 2024. arXiv:2411.04368 [cs]. → pages 4

[WLL+25] Suqing Wang, Zuchao Li, Shi Luohe, Bo Du, Hai Zhao, Yun Li, and Qianren Wang. From Parameters to Performance: A Data-Driven Study on LLM Structure and Development. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 26084–26101, Suzhou, China, November 2025. Association for Computational Linguistics. → pages 16

[WMZ+24] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhu Chen. MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark, November 2024. arXiv:2406.01574 [cs]. → pages 5

Appendix

Appendix A

Figures

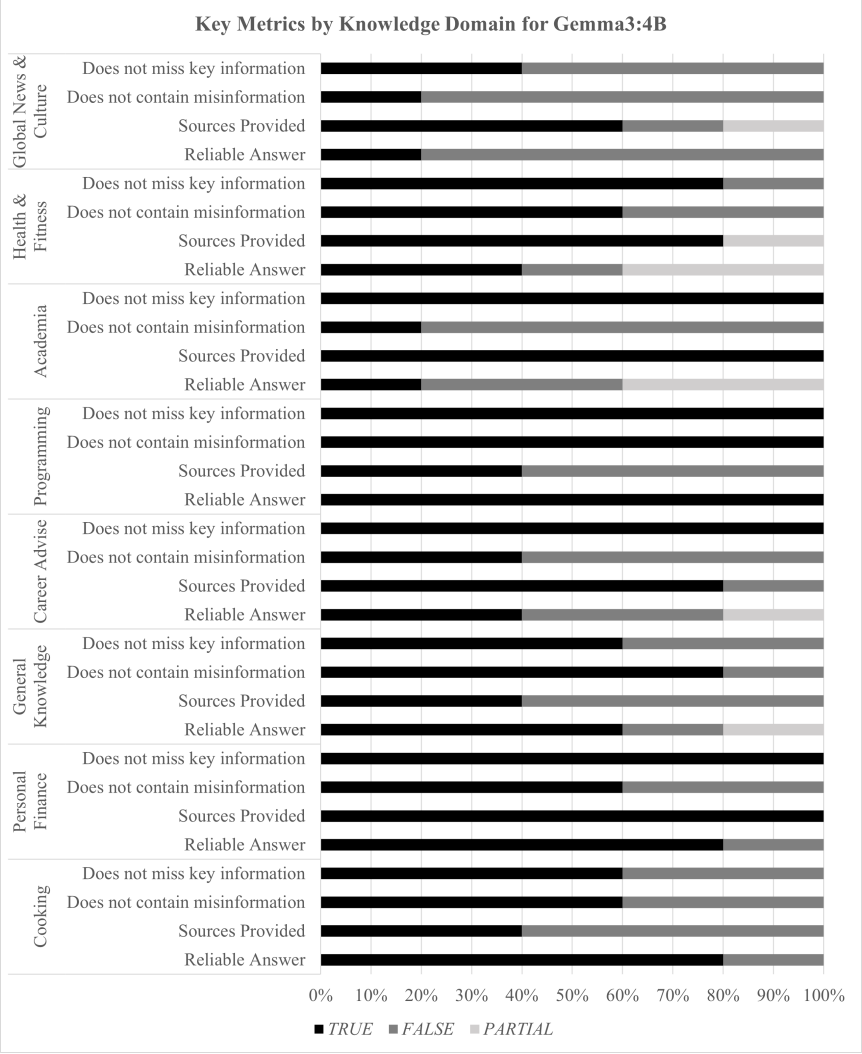


Figure A.1: Key metrics for *gemma3:4b* by knowledge-domain.

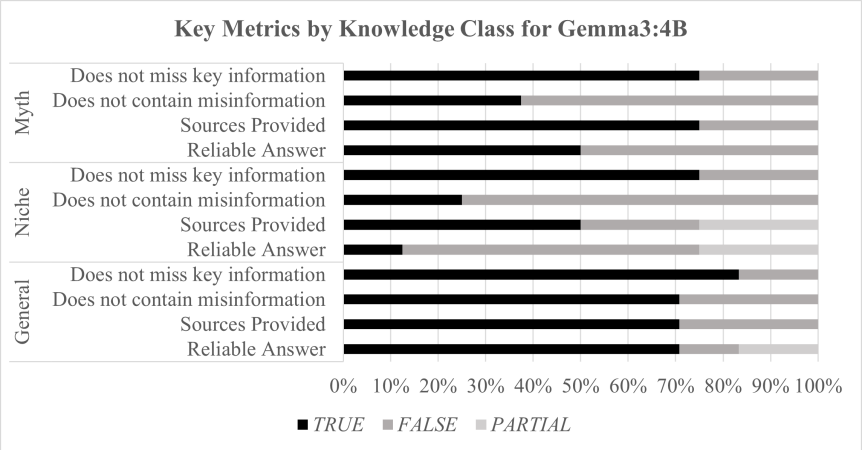


Figure A.2: Key metrics for *gemma3:4b* by knowledge-class.